Review notes · for you

Defenders: cryptography, networks, careers: unit check

Cyber security · Level 4 · Unit check

Time	Materials
10 min	Paper and a pen. No device needed.

Learning goals

- Fingerprints of data: hashing
- Locks and keys: public-key cryptography
- Networks and the defenders who watch them
- Careers in cyber security

Activity: Review: Defenders' quiz

1. Recall (3 min): Without notes, write one idea you remember from each of the four lessons.
2. Quiz (5 min): Take turns asking a partner one question per lesson. The partner answers and explains why.
3. Fix (2 min): Write one habit you will change, such as a unique passphrase for your email, or never "just looking" where you have no permission.

Work in pairs. Keep this short: the unit check comes next.

Discussion

- Why can't a good website send you your password?
- What does the padlock prove, and what doesn't it?
- What would you do if you found a weakness by accident?

Check yourself

- The idea that surprised me most, and one step I will take.

Try it this week
Pick the idea from this unit that surprised you most, explain it to someone at home, and write down one step you will take: to keep your own accounts safer, or towards a career in security.

Quiz sheet

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Name _____ Class _____ Date _____

1 **You hash a single word and then a whole film with SHA-256. How long is each fingerprint?**

- ◯ The film's is much longer than the word's
- ◯ Both are 256 bits: 64 hexadecimal characters
- ◯ As long as the input, but scrambled

2 **A website that stores only salted hashes can't email you your old password if you forget it.**

- ◯ True
- ◯ False

3 **Which statements about public and private keys are true?**

- ☐ Only your private key opens a message locked with your public key
- ☐ Anyone can check your signature with your public key
- ☐ Anyone can lock a message for you with your public key
- ☐ Send your private key to friends so they can write to you
- ☐ The HTTPS padlock proves a site is honest

4 **Put the steps of signing in to a well-built website in order.**

- ☐ It compares the result with the stored fingerprint
- ☐ If they match, you're signed in
- ☐ You type your password
- ☐ The site adds your salt and hashes what you typed

5 **Someone quietly changes the fees list on the university finance office's computer. Which security goal is broken?**

- ◯ Confidentiality
- ◯ Availability
- ◯ Integrity

6 **Match each layer of defence to its job.**

Firewall	• • • •	Record who connected, when and from where
Separate networks	• • • •	Decides which traffic may pass
Logs	• • • •	Keep things running when something fails
Backups and spare routes	• • • •	Keep the office apart from students' Wi-Fi

7 **Complete the sentence about RSA.**

RSA is safe because multiplying two huge _____ numbers is easy, but splitting the answer back into them is _____.

Word bank: extremely hard · quick · prime · even

8 **If one route through the internet breaks, packets can take another.**

- ◯ True
- ◯ False

9 **You notice your department's Wi-Fi admin password written on a noticeboard. What should you do?**

- ◯ Share it with friends so everyone sees the risk
- ◯ Use it to check the settings are safe
- ◯ Tell the IT team so it can be changed, and don't use it

10 **Ethical hackers may test any website they like, as long as they report what they find.**

- ◯ True
- ◯ False

Check your answers

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1. **Both are 256 bits: 64 hexadecimal characters**

   A hash always gives a fixed-size fingerprint, whatever goes in: for SHA-256, 256 bits, written as 64 hexadecimal characters.

2. **True**

   A hash can't be run backwards, so the site never has your password to send.

3. **Anyone can lock a message for you with your public key; Only your private key opens a message locked with your public key; Anyone can check your signature with your public key**

   The public key locks and checks; the private key opens and signs, so it never leaves you.

4. **Top to bottom: 3, 4, 1, 2**

   The site never needs your password itself: it repeats the same salted hash and compares fingerprints.

5. **Integrity**

   The data was changed without anyone noticing, so integrity was broken.

6. **Firewall — Decides which traffic may pass**
   Separate networks — Keep the office apart from students' Wi-Fi
   Logs — Record who connected, when and from where
   Backups and spare routes — Keep things running when something fails

   Each layer stops or catches a different problem, so if one fails, another still helps.

7. **RSA is safe because multiplying two huge prime numbers is easy, but splitting the answer back into them is extremely hard.**

   The whole trick is a one-way street in maths: multiplying primes takes a moment, while splitting a number hundreds of digits long back into its primes is beyond any computer today.

8. **True**

   There are many routes between two places, and routers send packets around a broken link.

9. **Tell the IT team so it can be changed, and don't use it**

   Using a password you weren't given is unauthorised access, however good the reason, and sharing it spreads the risk.

10. **False**

    Ethical hackers test only with the owner's written permission, and only within the agreed limits.